how much longer the game is *likely* to last. This assessment may become an important part of strategy. If I notice the banker fidget and look at his watch, I may well conjecture that the game is about to be brought to an end, and I may therefore feel tempted to defect. If I suspect that you too have noticed the banker fidgeting, I may fear that you too may be contemplating defection. I will probably be anxious to get my defection in first. Especially since I may fear that you are fearing that I …

The mathematician's simple distinction between the one-off Prisoner's Dilemma game and the Iterated Prisoner's Dilemma game is too simple. Each player can be expected to behave as if he possessed a continuously updated estimate of how long the game is likely to go on. The longer his estimate, the more he will play according to the mathematician's expectations for the true iterated game: in other words, the nicer, more forgiving, less envious he will be. The shorter his estimate of the future of the game, the more he will be inclined to play according to the mathematician's expectations for the one-off game: the nastier, and less forgiving will he be.

Axelrod draws a moving illustration of the importance of the shadow of the future from a remarkable phenomenon that grew up during the First World War, the so-called live-and-let-live system. His source is the research of the historian and sociologist Tony Ashworth. It is quite well known that at Christmas in 1914 British and German troops briefly fraternized and drank together in no man's land. Less well known, but in my opinion more interesting, is the fact that unofficial and unspoken nonaggression pacts, a 'live-and-let-live' system, flourished all up and down the front lines for at least two years starting in 1914. A senior British officer, on a visit to the trenches, is quoted as being astonished to observe German soldiers walking about within rifle range behind their own line. 'Our men appeared to take no notice. I privately made up my mind to do away with that sort of thing when we took over; such things should not be allowed. These people evidently did not know there was a war on. Both sides apparently believed in the policy of "live-and-let-live".'

The theory of games and the Prisoner's Dilemma had not been invented in those days but, with hindsight, we can see pretty clearly what was going on, and Axelrod provides a fascinating analysis. In the entrenched warfare of those times, the shadow of the future for each platoon was long. That is to say, each dug-in group of British soldiers